Motivation

- Leaky Integrate-and-Fire Neurons
    - work by integrating (accumulating) input currents over time to increase a "membrane potential" and "fire" (produce a spike) when this potential crosses a threshold, similar to how a leaky RC circuit charges. After firing, the potential resets. The "leaky" part refers to the fact that the membrane potential also gradually decays or "leaks" back to a resting value over time unless it is being driven by input
- The SCN idea
    - a population of LIF neurons only fire when it reduces the global representation error. This yields spare, irregular, yet coordinated spikes
    - The paper shows that the same network can implement a linear dynamical system (add a “slow” recurrent term).
    - Kalman filter for optimal state estimation through noisy observations
    - LQR for optimal control to a target z(t)
    - The control u is a linear readout of the spikes
    - -> Spiking LQG